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Prime CAT 05 2022 DILR

Scorecard (procreview.jsp?sid=aaaN5tjtX0b7WgArBjowyMon Jan 09 00:04:50 IST 2023&qsetId=U3YFbta7zl4=&qsetName=Prime CAT 05 2022 DILR)

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Qs Analysis (QsAnalysis.jsp?sid=aaaN5tjtX0b7WgArBjowyMon Jan 09 00:04:50 IST 2023&qsetId=U3YFbta7zl4=&qsetName=Prime CAT 05 2022 DILR)

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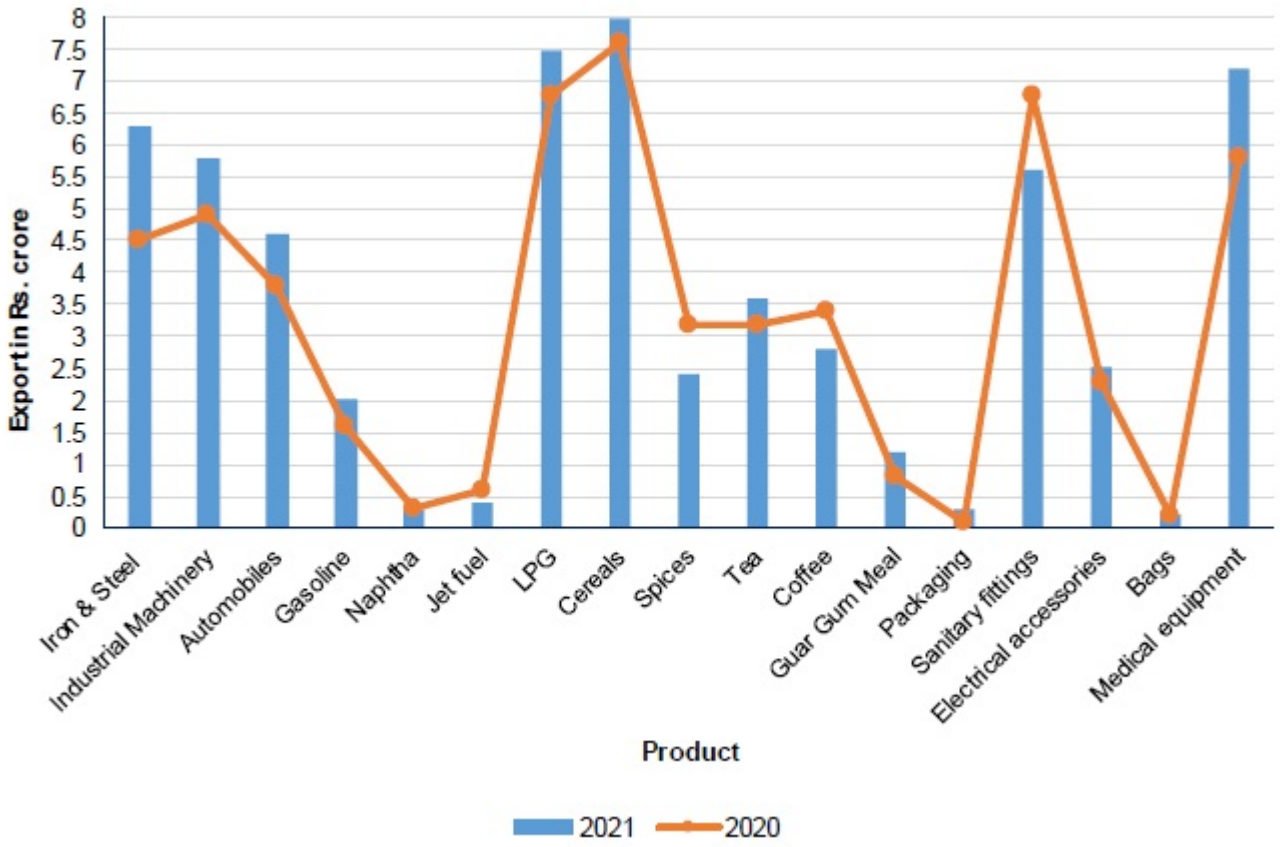
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Section-1

Sec 1

Directions for questions 1 to 6: Answer the questions on the basis of the information given below.

The bar graph given below shows the total revenue (in Rs. crore) from exports of various products by a company ABC Private Limited in 2021. The products are categorized under four different heads. Iron & Steel, Industrial Machinery, Automobiles come under Engineering Products. Gasoline, Naphtha, Jet fuel, LPG come under Petroleum Products. Cereals, Spices, Tea, Coffee, Guar Gum Meal come under Agricultural Products and the remaining five come under Plastic & Linoleum Products. The line graph in the chart shows the revenue (in Rs. crore) from exports of the corresponding products in 2020.



Q.1 [11831809]

The number of products whose exports declined from 2020 to 2021 is

Solution:

Correct Answer : 4

Answer key/Solution

From the given information, the table can be shown as:

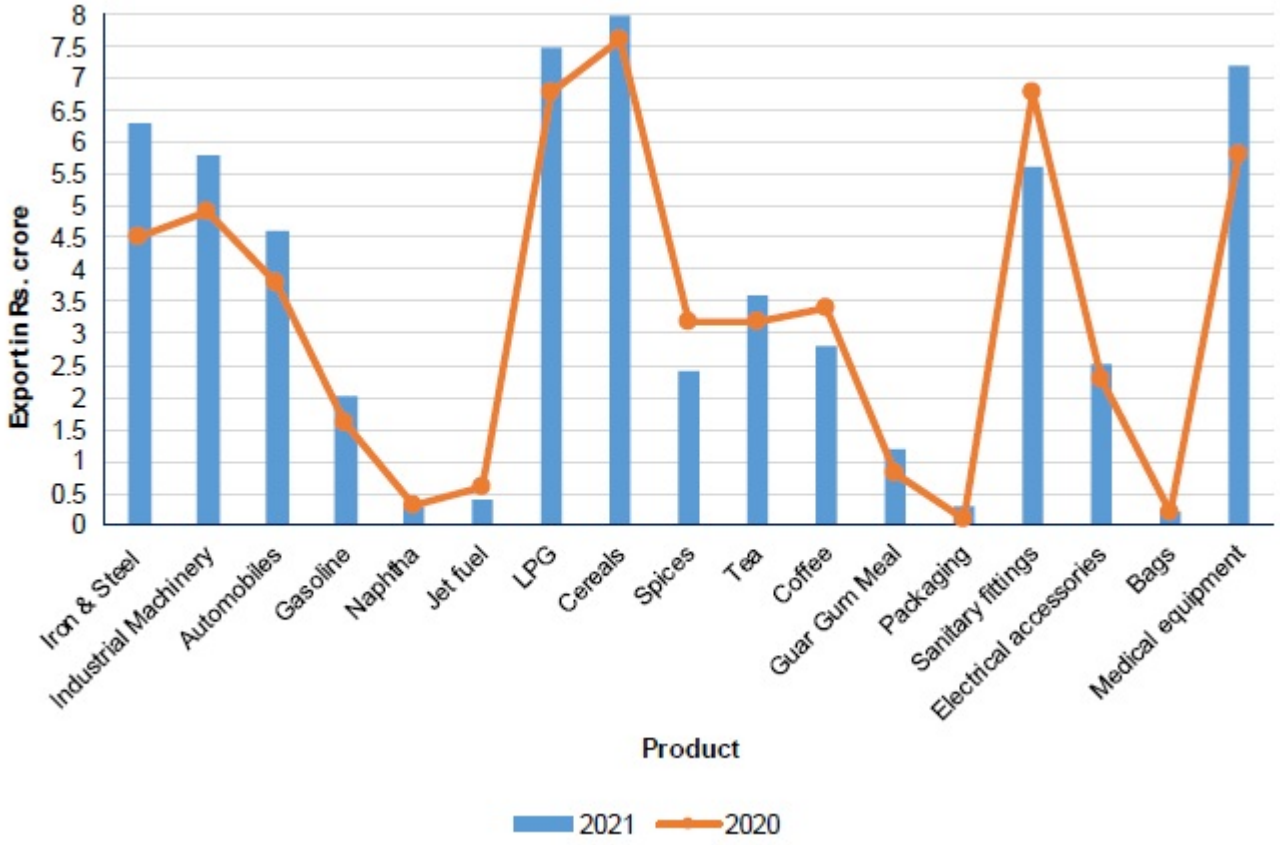
	Product	2021	2020	Increase/Decrease
Engineering Products	Iron & Steel	6.3	4.5	1.8
	Industrial Machinery	5.8	4.9	0.9
	Automobiles	4.6	3.8	0.8
		16.7	**13.2**	
Petroleum Products	Gasoline	2	1.6	0.4
	Naphtha	0.3	0.3	0
	Jet fuel	0.4	0.6	-0.2
	LPG	7.5	6.8	0.7
		10.2	**9.3**	
Agricultural Products	Cereals	8	7.6	0.4
	Spices	2.4	3.2	-0.8
	Tea	3.6	3.2	0.4
	Coffee	2.8	3.4	-0.6
	Guar Gum Meal	1.2	0.8	0.4
		18	**18.2**	
Plastic and Linoleum Products	Packaging	0.3	0.1	0.2
	Sanitary fittings	5.6	6.8	-1.2
	Electrical accessories	2.5	2.3	0.2
	Bags	0.2	0.2	0
	Medical equipment	7.2	5.8	1.4
		15.8	**15.2**	
		60.7	55.9	

There are 4 products (Jet fuel, Spices, Coffee and Sanitary fittings) whose exports declined from 2020 to 2021.

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Q.2 [11831809]

Which of the following products shows highest increase in exports from 2020 to 2021?

1 ◯ Iron & Steel

2 ◯ Industrial Machinery

3 ◯ Automobiles

4 ◯ Medical equipment

Solution:

Correct Answer : 1

Answer key/Solution

From the given information, the table can be shown as:

	Product	2021	2020	Increase/Decrease
Engineering Products	Iron & Steel	6.3	4.5	1.8
	Industrial Machinery	5.8	4.9	0.9
	Automobiles	4.6	3.8	0.8
		16.7	**13.2**	
Petroleum Products	Gasoline	2	1.6	0.4
	Naphtha	0.3	0.3	0
	Jet fuel	0.4	0.6	-0.2
	LPG	7.5	6.8	0.7
		10.2	**9.3**	
Agricultural Products	Cereals	8	7.6	0.4
	Spices	2.4	3.2	-0.8
	Tea	3.6	3.2	0.4
	Coffee	2.8	3.4	-0.6
	Guar Gum Meal	1.2	0.8	0.4
		18	**18.2**	
Plastic and Linoleum Products	Packaging	0.3	0.1	0.2
	Sanitary fittings	5.6	6.8	-1.2
	Electrical accessories	2.5	2.3	0.2
	Bags	0.2	0.2	0
	Medical equipment	7.2	5.8	1.4
		15.8	**15.2**	
		60.7	55.9	

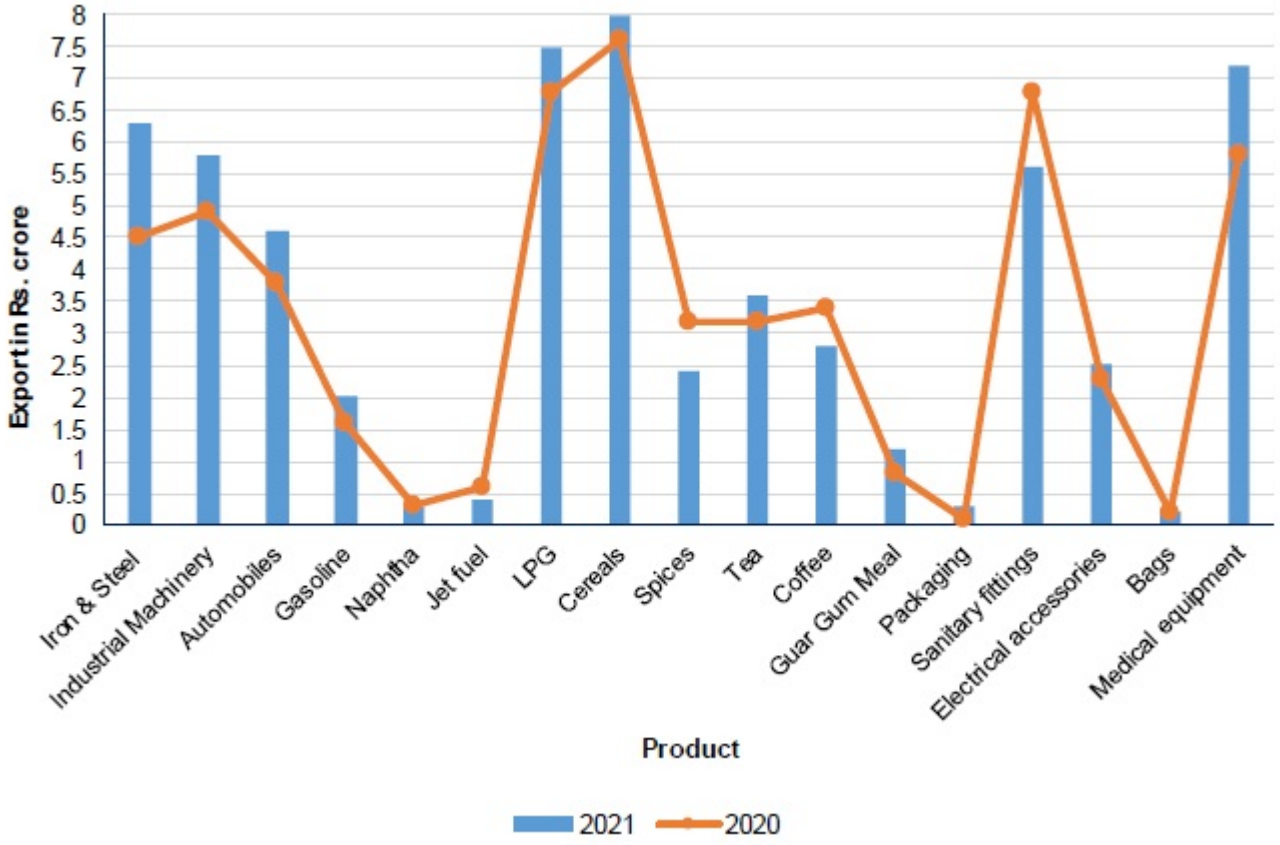
Increase in exports of Iron and Steel = 6.3 – 4.5 = Rs. 1.8 crores

Hence, Iron & Steel shows the highest increase in exports from 2020 to 2021.

Bookmark FeedBack

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Q.3 [11831809]

The revenue (in Rs. crore) from the export of Plastic & Linoleum Products in 2020 is how much more than the revenue from the export of Petroleum Products in the same year?

1 ◯ 4.9

2 ◯ 7.9

3 ◯ 9.5

4 ◯ 5.9

Solution:

Correct Answer : 4

Answer key/Solution

From the given information, the table can be shown as:

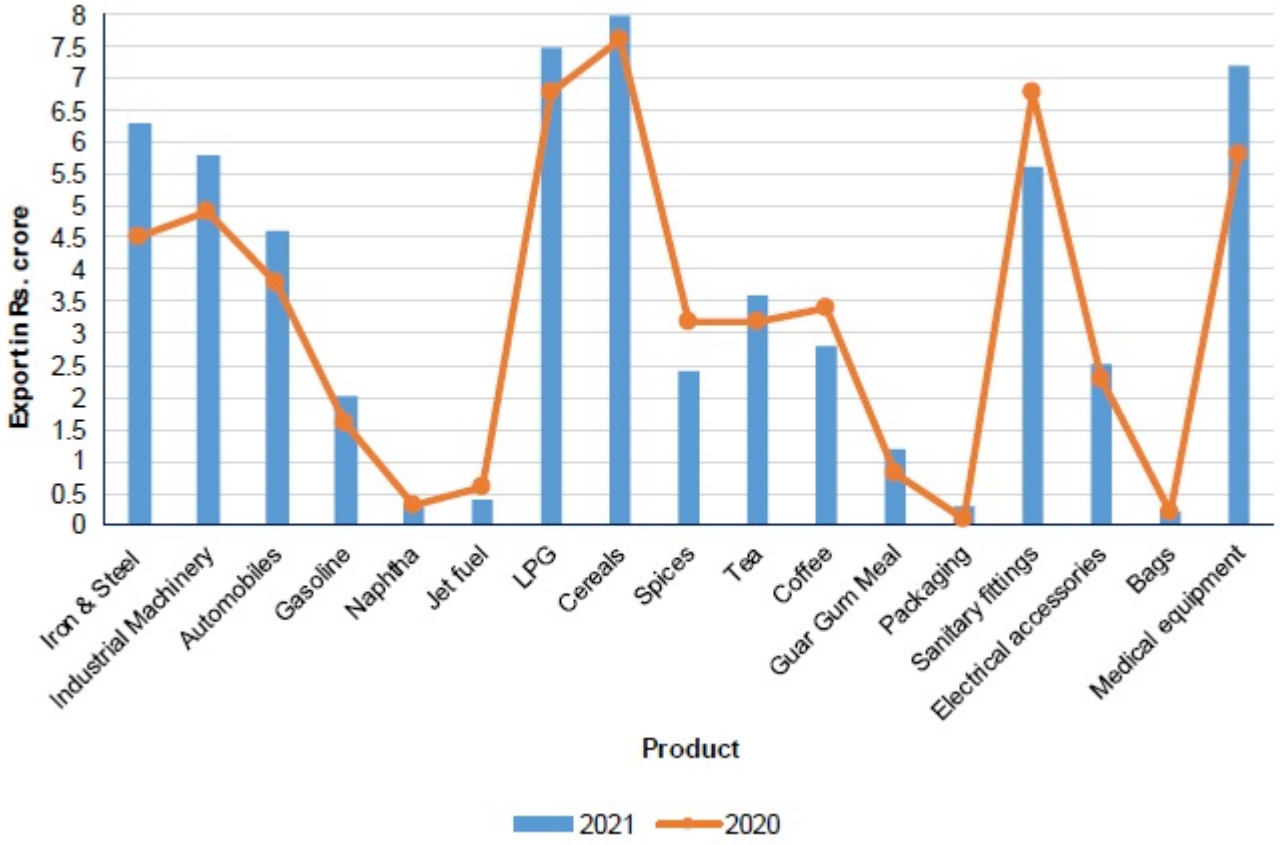
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	Industrial Machinery	5.8	4.9	0.9
	Automobiles	4.6	3.8	0.8
		16.7	**13.2**	
Petroleum Products	Gasoline	2	1.6	0.4
	Naphtha	0.3	0.3	0
	Jet fuel	0.4	0.6	-0.2
	LPG	7.5	6.8	0.7
		10.2	**9.3**	
Agricultural Products	Cereals	8	7.6	0.4
	Spices	2.4	3.2	-0.8
	Tea	3.6	3.2	0.4
	Coffee	2.8	3.4	-0.6
	Guar Gum Meal	1.2	0.8	0.4
		18	**18.2**	
Plastic and Linoleum Products	Packaging	0.3	0.1	0.2
	Sanitary fittings	5.6	6.8	-1.2
	Electrical accessories	2.5	2.3	0.2
	Bags	0.2	0.2	0
	Medical equipment	7.2	5.8	1.4
		15.8	**15.2**	
		60.7	55.9	

Required export = 15.2 – 9.3 = Rs. 5.9 crore.

Bookmark FeedBack

Directions for questions 1 to 6: Answer the questions on the basis of the information given below.

The bar graph given below shows the total revenue (in Rs. crore) from exports of various products by a company ABC Private Limited in 2021. The products are categorized under four different heads. Iron & Steel, Industrial Machinery, Automobiles come under Engineering Products. Gasoline, Naphtha, Jet fuel, LPG come under Petroleum Products. Cereals, Spices, Tea, Coffee, Guar Gum Meal come under Agricultural Products and the remaining five come under Plastic & Linoleum Products. The line graph in the chart shows the revenue (in Rs. crore) from exports of the corresponding products in 2020.



Q.4 [11831809]

What is the approximate percentage growth in the total exports from 2020 to 2021?

1 ◯ 4.5%

2 ◯ 8.5%

3 ◯ 16.5%

4 ◯ 20.5%

Solution:
Correct Answer : 2

Answer key/Solution

From the given information, the table can be shown as:

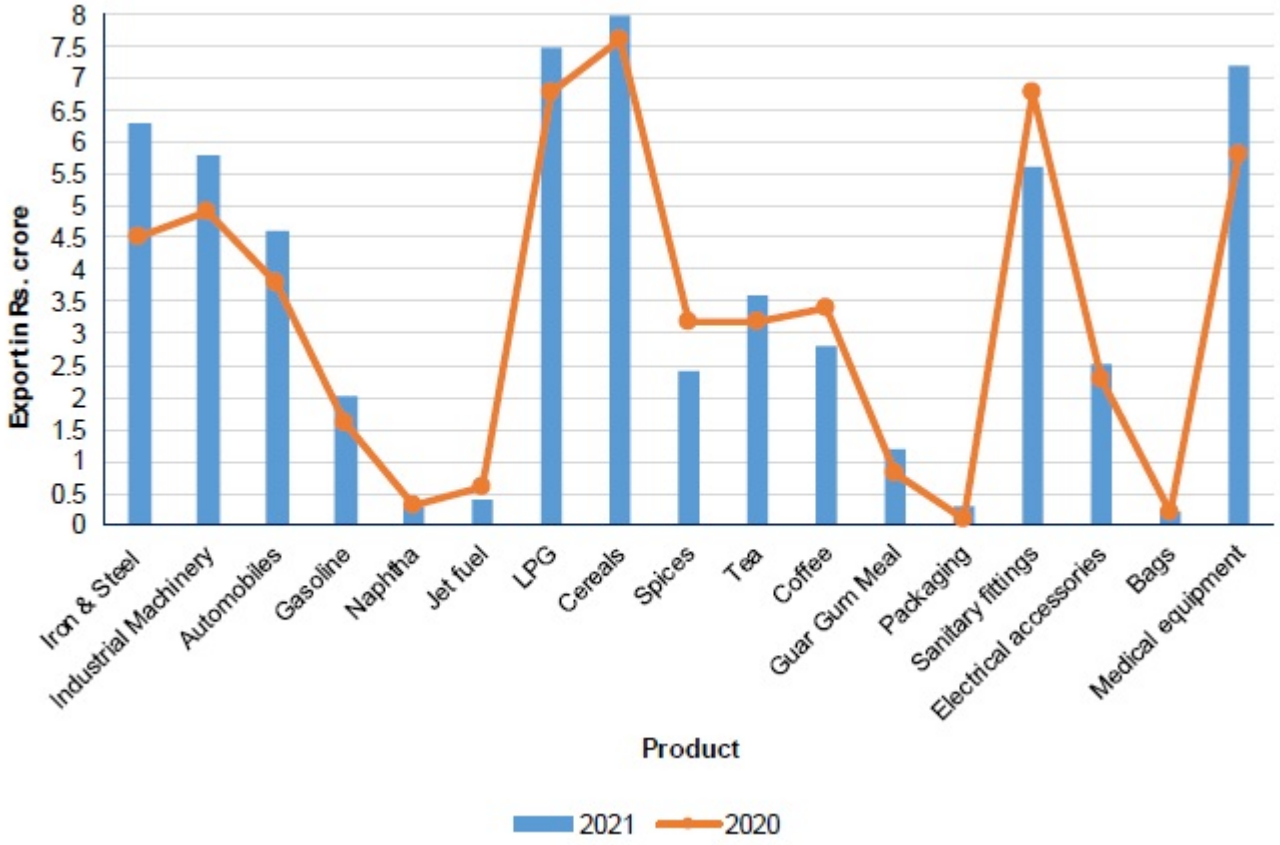
	Product	2021	2020	Increase/Decrease
Engineering Products	Iron & Steel	6.3	4.5	1.8
	Industrial Machinery	5.8	4.9	0.9
	Automobiles	4.6	3.8	0.8
		16.7	**13.2**	
Petroleum Products	Gasoline	2	1.6	0.4
	Naphtha	0.3	0.3	0
	Jet fuel	0.4	0.6	-0.2
	LPG	7.5	6.8	0.7
		10.2	**9.3**	
Agricultural Products	Cereals	8	7.6	0.4
	Spices	2.4	3.2	-0.8
	Tea	3.6	3.2	0.4
	Coffee	2.8	3.4	-0.6
	Guar Gum Meal	1.2	0.8	0.4
		18	**18.2**	
Plastic and Linoleum Products	Packaging	0.3	0.1	0.2
	Sanitary fittings	5.6	6.8	-1.2
	Electrical accessories	2.5	2.3	0.2
	Bags	0.2	0.2	0
	Medical equipment	7.2	5.8	1.4
		15.8	**15.2**	
		60.7	**55.9**	

Required percentage growth in the total export from 2020 to 2021 = $\frac{60.7 - 55.9}{55.9} \times 100 \approx 8.5\%$

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Directions for questions 1 to 6: Answer the questions on the basis of the information given below.

The bar graph given below shows the total revenue (in Rs. crore) from exports of various products by a company ABC Private Limited in 2021. The products are categorized under four different heads. Iron & Steel, Industrial Machinery, Automobiles come under Engineering Products. Gasoline, Naphtha, Jet fuel, LPG come under Petroleum Products. Cereals, Spices, Tea, Coffee, Guar Gum Meal come under Agricultural Products and the remaining five come under Plastic & Linoleum Products. The line graph in the chart shows the revenue (in Rs. crore) from exports of the corresponding products in 2020.



Q.5 [11831809]

If Plastic & Linoleum Products, Agricultural Products, Petroleum Products and Engineering Products are represented by category 1, 2, 3 and 4 respectively, then which category shows the highest percentage growth in exports from 2020 to 2021?

Solution:

Correct Answer : 4

Answer key/Solution

From the given information, the table can be shown as:

	Product	2021	2020	Increase/Decrease
Engineering Products	Iron & Steel	6.3	4.5	1.8
	Industrial Machinery	5.8	4.9	0.9
	Automobiles	4.6	3.8	0.8
		16.7	**13.2**	
Petroleum Products	Gasoline	2	1.6	0.4
	Naphtha	0.3	0.3	0
	Jet fuel	0.4	0.6	-0.2
	LPG	7.5	6.8	0.7
		10.2	**9.3**	
Agricultural Products	Cereals	8	7.6	0.4
	Spices	2.4	3.2	-0.8
	Tea	3.6	3.2	0.4
	Coffee	2.8	3.4	-0.6
	Guar Gum Meal	1.2	0.8	0.4
		18	**18.2**	
Plastic and Linoleum Products	Packaging	0.3	0.1	0.2
	Sanitary fittings	5.6	6.8	-1.2
	Electrical accessories	2.5	2.3	0.2
	Bags	0.2	0.2	0
	Medical equipment	7.2	5.8	1.4
		15.8	**15.2**	
		60.7	**55.9**	

The percentage growth for:

Engineering products $= \frac{16.7 - 13.2}{13.2} \times 100 \approx 26.5\%$

Petroleum Products $= \frac{10.2 - 9.3}{9.3} \times 100 \approx 9.7\%$

Agricultural Products = No growth

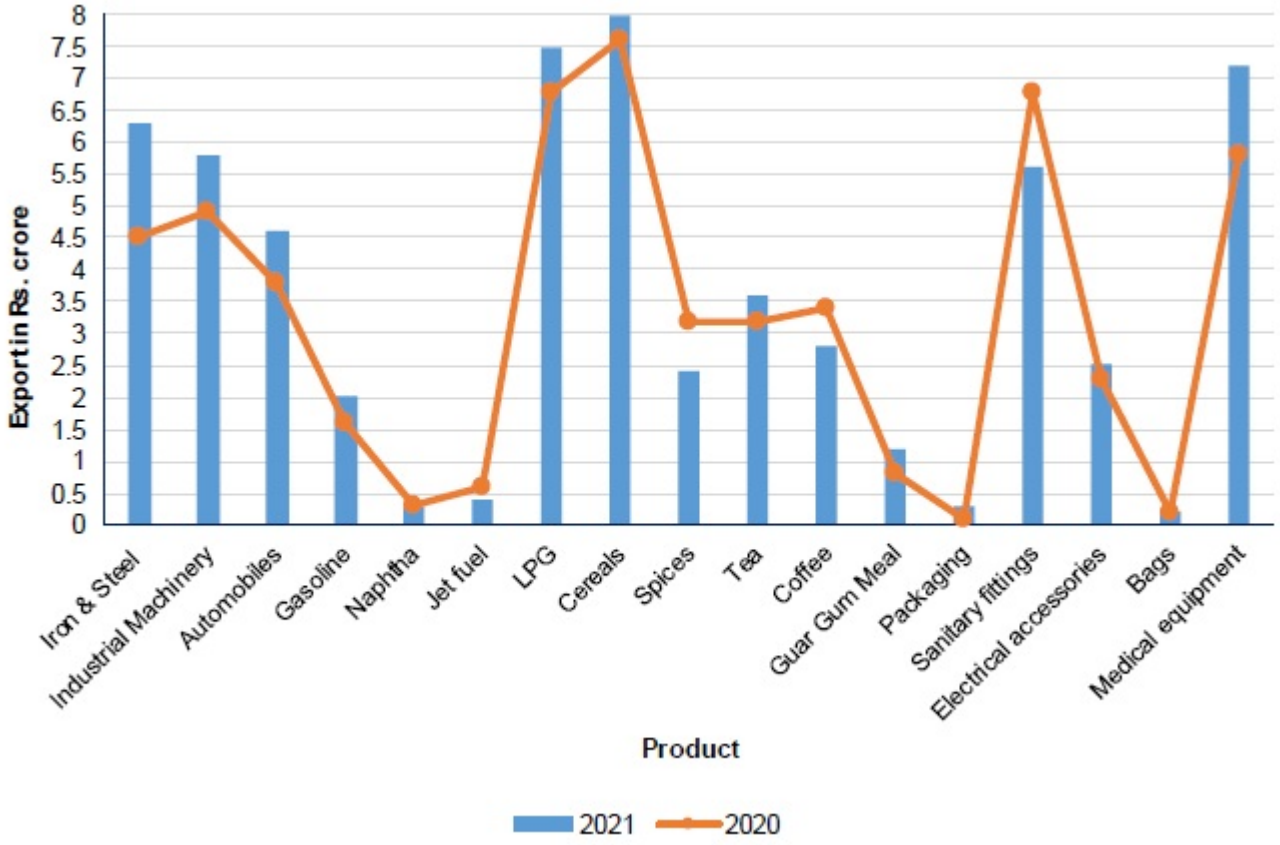
Plastic & Linoleum Products $= \frac{15.8 - 15.2}{15.2} \times 100 \approx 4\%$

Hence, highest percentage growth is in Engineering Products i.e., category 4.

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Directions for questions 1 to 6: Answer the questions on the basis of the information given below.

The bar graph given below shows the total revenue (in Rs. crore) from exports of various products by a company ABC Private Limited in 2021. The products are categorized under four different heads. Iron & Steel, Industrial Machinery, Automobiles come under Engineering Products. Gasoline, Naphtha, Jet fuel, LPG come under Petroleum Products. Cereals, Spices, Tea, Coffee, Guar Gum Meal come under Agricultural Products and the remaining five come under Plastic & Linoleum Products. The line graph in the chart shows the revenue (in Rs. crore) from exports of the corresponding products in 2020.



Q.6 [11831809]

Among the products that have witnessed an increase in revenue from 2020 to 2021, the percentage increase in revenue from 2020 to 2021 is calculated for those that have increased by less than Rs. 0.5 crore and then they are ranked in order of the percentage increase. Which of the following options shows the correct order of the products on this basis?

1 ◯ Packaging, Gaur Gum Meal, Gasoline, Tea

2 ◯ Gasoline, Tea, Gaur Gum Meal, Packaging

3 ◯ Tea, Gasoline, Gaur Gum Meal, Packaging

4 ◯ Tea, Gasoline, Packaging, Gaur Gum Meal

Solution:
Correct Answer : 3

Answer key/Solution

From the given information, the table can be shown as:

	Product	2021	2020	Increase/Decrease
Engineering Products	Iron & Steel	6.3	4.5	1.8
	Industrial Machinery	5.8	4.9	0.9
	Automobiles	4.6	3.8	0.8
		16.7	**13.2**	
Petroleum Products	Gasoline	2	1.6	0.4
	Naphtha	0.3	0.3	0
	Jet fuel	0.4	0.6	-0.2
	LPG	7.5	6.8	0.7
		10.2	**9.3**	
Agricultural Products	Cereals	8	7.6	0.4
	Spices	2.4	3.2	-0.8
	Tea	3.6	3.2	0.4
	Coffee	2.8	3.4	-0.6
	Guar Gum Meal	1.2	0.8	0.4
		18	**18.2**	
Plastic and Linoleum Products	Packaging	0.3	0.1	0.2
	Sanitary fittings	5.6	6.8	-1.2
	Electrical accessories	2.5	2.3	0.2
	Bags	0.2	0.2	0
	Medical equipment	7.2	5.8	1.4
		15.8	**15.2**	
		60.7	**55.9**	

Among the products that have witnessed an increase in revenue by less than Rs. 0.5 crore from 2020 to 2021, the products mentioned in the options are Gasoline, Cereals, Tea, Gaur Gum Meal, Packaging and Electrical accessories.
The percentage increase in revenue from 2020 to 2021 for these products is:
Tea = 0.4/3.2 × 100 = 12.5%
Gasoline = 0.4/1.6 × 100 = 25%
Gaur Gum Meal = 0.4/0.8 × 100 = 50%
Packaging = 0.2/0.1 × 100 = 200%.
Hence, answer (3) is correct.

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Directions for questions 7 to 10: Answer the questions on the basis of the information given below.

A team of 21 teachers was appointed to teach in 16 villages of a particular district under "Project Buniyad", during a span of 4 months. Out of these 16 villages, 4 villages are "Under-developed", while the remaining are "Developed". Teaching in each Under-developed village was completed in a different month. The teachers are divided into four teams - P1, P2, P3 and P4 - for this purpose and each team covers one village per month. There are 10 Post-Graduate teachers (PD) and the remaining are Graduate teachers (GD). Teams P1, P2, P3, P4 are allotted one Under-developed village each in the first, second, third and fourth month, respectively. Each month, the team allotted the Under-developed village has one more teacher than the remaining teams. In the first month, P1 has one more PD than P2, P2 has one more PD than P3, P3 has one more PD than P4.

Between two successive months, the composition of the teams changes as follows:
(i) The team allotted the under-developed village gets two PDs from the team which was allotted the Under-developed village in the previous month. In exchange, one GD is shifted from the former team to the latter team.
(ii) After the above exchange, if P1 has any PDs left and P4 has any GDs left, then one PD is shifted from P1 to P4, and one GD is shifted from P4 to P1. Also, if P2 has any PD and P3 has any GD, then one PD is shifted from P2 to P3, and one GD is shifted from P3 to P2.
(iii) The total number of students taught in the four months was 270, 274, 273 and 271 in that order. In every month, starting from the first month, a team with a higher number of PDs taught higher number of students. If for any two teams, the number of PDs were equal, then the team with higher number of GDs taught higher number of students. It was observed that no team taught more than 70 students and less than 66 students.

Q.7 [11831809]
The composition of team P2 in the first month is same as composition of team P_____ in the third month.

Solution:
Correct Answer : 4

Answer key/Solution

Step 1:
Given that there are 10 PD and 11 GD. In the first month, since P1 has one more PD than P2, who in turn has one more PD than P3, ... till P4, the number of PDs in P1, P2, P3, and P4 must be 4, 3, 2, and 1. Also, the team that is allotted the under-developed village has one more teacher than the rest. Hence, the team that is assigned the under-developed village will have 6 teachers, while the other teams will have 5 teachers. Since P1 is assigned the under-developed village for the first month, P1 will have 6 teachers, and the other teams will have 5 teachers each.
The number of students taught in first month was 270 = 66 + 67 + 68 + 69. Since greater number of PDs is directly proportional to the number of students taught, therefore,

Team	PD	GD	Total	Students
P1	4	2	6	69
P2	3	2	5	68
P3	2	3	5	67
P4	1	4	5	66

Step 2:
In the second month, P2 will be allotted the under-developed village. From condition (i), two PDs will be transferred from P1 to P2. One GD is transferred from P2 to P1. From condition (ii), one PD will be transferred from P1 to P4, one GD will be transferred from P4 to P1. Similar transfers will happen between P2 and P3. The following table provides the number of teachers in each team in the second month:

Teams	PD	GD	Total	Students
P1	1	4	5	67
P2	4	2	6	70
P3	3	2	5	69
P4	2	3	5	68

Step 3:
The following table provides the number of teachers in each team in the third month:

Teams	PD	GD	Total	Students
P1	0	5	5	66
P2	1	4	5	68
P3	6	0	6	70
P4	3	2	5	69

Step 4:
From condition (ii), one PD must be transferred from P1 to P4. However, since there are no PDs in P1, this will not happen. Also, one PD must be transferred from P2 to P3 and one GD must be transferred from P3 to P2. The following table provides the number of teachers in each team in the fourth month:

Teams	PD	GD	Total	Students
P1	0	5	5	66/67
P2	0	5	5	67/66
P3	5	0	5	68
P4	5	1	6	70

The composition of team P2 in the first month is same as the composition of team P4 in the third month.

Bookmark FeedBack

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Between two successive months, the composition of the teams changes as follows:
(i) The team allotted the under-developed village gets two PDs from the team which was allotted the Under-developed village in the previous month. In exchange, one GD is shifted from the former team to the latter team.
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(iii) The total number of students taught in the four months was 270, 274, 273 and 271 in that order. In every month, starting from the first month, a team with a higher number of PDs taught higher number of students. If for any two teams, the number of PDs were equal, then the team with higher number of GDs taught higher number of students. It was observed that no team taught more than 70 students and less than 66 students.

Q.8 [11831809]
The number of PD in P1 and P2 for the projects in the fourth month are, respectively:

1 ◯ (0, 0)

2 ◯ (5, 5)

3 ◯ (0, 5)

4 ◯ (5, 0)

Solution:
Correct Answer : 1

Answer key/Solution

Step 1:
Given that there are 10 PD and 11 GD. In the first month, since P1 has one more PD than P2, who in turn has one more PD than P3, ... till P4, the number of PDs in P1, P2, P3, and P4 must be 4, 3, 2, and 1. Also, the team that is allotted the under-developed village has one more teacher than the rest. Hence, the team that is assigned the under-developed village will have 6 teachers, while the other teams will have 5 teachers. Since P1 is assigned the under-developed village for the first month, P1 will have 6 teachers, and the other teams will have 5 teachers each.
The number of students taught in first month was 270 = 66 + 67 + 68 + 69. Since greater number of PDs is directly proportional to the number of students taught, therefore,

Team	PD	GD	Total	Students
P1	4	2	6	69
P2	3	2	5	68
P3	2	3	5	67
P4	1	4	5	66

Step 2:
In the second month, P2 will be allotted the under-developed village. From condition (i), two PDs will be transferred from P1 to P2. One GD is transferred from P2 to P1. From condition (ii), one PD will be transferred from P1 to P4, one GD will be transferred from P4 to P1. Similar transfers will happen between P2 and P3. The following table provides the number of teachers in each team in the second month:

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P3	3	2	5	69
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Step 3:
The following table provides the number of teachers in each team in the third month:

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P3	6	0	6	70
P4	3	2	5	69

Step 4:
From condition (ii), one PD must be transferred from P1 to P4. However, since there are no PDs in P1, this will not happen. Also, one PD must be transferred from P2 to P3 and one GD must be transferred from P3 to P2. The following table provides the number of teachers in each team in the fourth month:

Teams	PD	GD	Total	Students
P1	0	5	5	66/67
P2	0	5	5	67/66
P3	5	0	5	68
P4	5	1	6	70

The number of PD in P1 and P2 for the projects in the fourth month are 0 and 0 respectively.

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Directions for questions 7 to 10: Answer the questions on the basis of the information given below.

A team of 21 teachers was appointed to teach in 16 villages of a particular district under "Project Buniyad", during a span of 4 months. Out of these 16 villages, 4 villages are "Under-developed", while the remaining are "Developed". Teaching in each Under-developed village was completed in a different month. The teachers are divided into four teams - P1, P2, P3 and P4 - for this purpose and each team covers one village per month. There are 10 Post-Graduate teachers (PD) and the remaining are Graduate teachers (GD). Teams P1, P2, P3, P4 are allotted one Under-developed village each in the first, second, third and fourth month, respectively. Each month, the team allotted the Under-developed village has one more teacher than the remaining teams. In the first month, P1 has one more PD than P2, P2 has one more PD than P3, P3 has one more PD than P4.

Between two successive months, the composition of the teams changes as follows:
(i) The team allotted the under-developed village gets two PDs from the team which was allotted the Under-developed village in the previous month. In exchange, one GD is shifted from the former team to the latter team.
(ii) After the above exchange, if P1 has any PDs left and P4 has any GDs left, then one PD is shifted from P1 to P4, and one GD is shifted from P4 to P1. Also, if P2 has any PD and P3 has any GD, then one PD is shifted from P2 to P3, and one GD is shifted from P3 to P2.
(iii) The total number of students taught in the four months was 270, 274, 273 and 271 in that order. In every month, starting from the first month, a team with a higher number of PDs taught higher number of students. If for any two teams, the number of PDs were equal, then the team with higher number of GDs taught higher number of students. It was observed that no team taught more than 70 students and less than 66 students.

Q.9 [11831809]
Total how many students were taught by team P4 in four months?

1 ◯ 273

2 ◯ 270

3 ◯ 274

4 ◯ Cannot be determined.

Solution:
Correct Answer : 1

Answer key/Solution

Step 1:
Given that there are 10 PD and 11 GD. In the first month, since P1 has one more PD than P2, who in turn has one more PD than P3, ... till P4, the number of PDs in P1, P2, P3, and P4 must be 4, 3, 2, and 1. Also, the team that is allotted the under-developed village has one more teacher than the rest. Hence, the team that is assigned the under-developed village will have 6 teachers, while the other teams will have 5 teachers. Since P1 is assigned the under-developed village for the first month, P1 will have 6 teachers, and the other teams will have 5 teachers each.
The number of students taught in first month was 270 = 66 + 67 + 68 + 69. Since greater number of PDs is directly proportional to the number of students taught, therefore,

Team	PD	GD	Total	Students
P1	4	2	6	69
P2	3	2	5	68
P3	2	3	5	67
P4	1	4	5	66

Step 2:
In the second month, P2 will be allotted the under-developed village. From condition (i), two PDs will be transferred from P1 to P2. One GD is transferred from P2 to P1. From condition (ii), one PD will be transferred from P1 to P4, one GD will be transferred from P4 to P1. Similar transfers will happen between P2 and P3. The following table provides the number of teachers in each team in the second month:

Teams	PD	GD	Total	Students
P1	1	4	5	67
P2	4	2	6	70
P3	3	2	5	69
P4	2	3	5	68

Step 3:
The following table provides the number of teachers in each team in the third month:

Teams	PD	GD	Total	Students
P1	0	5	5	66
P2	1	4	5	68
P3	6	0	6	70
P4	3	2	5	69

Step 4:
From condition (ii), one PD must be transferred from P1 to P4. However, since there are no PDs in P1, this will not happen. Also, one PD must be transferred from P2 to P3 and one GD must be transferred from P3 to P2. The following table provides the number of teachers in each team in the fourth month:

Teams	PD	GD	Total	Students
P1	0	5	5	66/67
P2	0	5	5	67/66
P3	5	0	5	68
P4	5	1	6	70

273 students were taught by team P4 in four months.

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Directions for questions 7 to 10: Answer the questions on the basis of the information given below.

A team of 21 teachers was appointed to teach in 16 villages of a particular district under "Project Buniyad", during a span of 4 months. Out of these 16 villages, 4 villages are "Under-developed", while the remaining are "Developed". Teaching in each Under-developed village was completed in a different month. The teachers are divided into four teams - P1, P2, P3 and P4 - for this purpose and each team covers one village per month. There are 10 Post-Graduate teachers (PD) and the remaining are Graduate teachers (GD). Teams P1, P2, P3, P4 are allotted one Under-developed village each in the first, second, third and fourth month, respectively. Each month, the team allotted the Under-developed village has one more teacher than the remaining teams. In the first month, P1 has one more PD than P2, P2 has one more PD than P3, P3 has one more PD than P4.

Between two successive months, the composition of the teams changes as follows:
(i) The team allotted the under-developed village gets two PDs from the team which was allotted the Under-developed village in the previous month. In exchange, one GD is shifted from the former team to the latter team.
(ii) After the above exchange, if P1 has any PDs left and P4 has any GDs left, then one PD is shifted from P1 to P4, and one GD is shifted from P4 to P1. Also, if P2 has any PD and P3 has any GD, then one PD is shifted from P2 to P3, and one GD is shifted from P3 to P2.
(iii) The total number of students taught in the four months was 270, 274, 273 and 271 in that order. In every month, starting from the first month, a team with a higher number of PDs taught higher number of students. If for any two teams, the number of PDs were equal, then the team with higher number of GDs taught higher number of students. It was observed that no team taught more than 70 students and less than 66 students.

Q.10 [11831809]
Rohit, a PD teacher under project Buniyaad earned a compensation of Rs. 900 per student he taught in four months. If Rohit was always a part of one team and earned the maximum possible compensation, then he was the part of team P_____.

1 ◯ 1

2 ◯ 2

3 ◯ 3

4 ◯ 4

Solution:
Correct Answer : 3

Answer key/Solution

Step 1:
Given that there are 10 PD and 11 GD. In the first month, since P1 has one more PD than P2, who in turn has one more PD than P3, ... till P4, the number of PDs in P1, P2, P3, and P4 must be 4, 3, 2, and 1. Also, the team that is allotted the under-developed village has one more teacher than the rest. Hence, the team that is assigned the under-developed village will have 6 teachers, while the other teams will have 5 teachers. Since P1 is assigned the under-developed village for the first month, P1 will have 6 teachers, and the other teams will have 5 teachers each.
The number of students taught in first month was 270 = 66 + 67 + 68 + 69. Since greater number of PDs is directly proportional to the number of students taught, therefore,

Team	PD	GD	Total	Students
P1	4	2	6	69
P2	3	2	5	68
P3	2	3	5	67
P4	1	4	5	66

Step 2:
In the second month, P2 will be allotted the under-developed village. From condition (i), two PDs will be transferred from P1 to P2. One GD is transferred from P2 to P1. From condition (ii), one PD will be transferred from P1 to P4, one GD will be transferred from P4 to P1. Similar transfers will happen between P2 and P3. The following table provides the number of teachers in each team in the second month:

Teams	PD	GD	Total	Students
P1	1	4	5	67
P2	4	2	6	70
P3	3	2	5	69
P4	2	3	5	68

Step 3:
The following table provides the number of teachers in each team in the third month:

Teams	PD	GD	Total	Students
P1	0	5	5	66
P2	1	4	5	68
P3	6	0	6	70
P4	3	2	5	69

Step 4:
From condition (ii), one PD must be transferred from P1 to P4. However, since there are no PDs in P1, this will not happen. Also, one PD must be transferred from P2 to P3 and one GD must be transferred from P3 to P2. The following table provides the number of teachers in each team in the fourth month:

Teams	PD	GD	Total	Students
P1	0	5	5	66/67
P2	0	5	5	67/66
P3	5	0	5	68
P4	5	1	6	70

Compensation earned by Rohit if he was a part of team:
P1 = 900 × (69 + 67) = Rs.900 × 136

P2 = 900 × (68 + 70 + 68) = Rs.900 × 206
P3 = 900 × (67 + 69 + 70 + 68) = Rs.900 × 274
P4 = 900 × (66 + 68 + 69 + 70) = Rs.900 × 273
Without actual calculation, it can be observed that P3 will give the maximum possible compensation.

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Directions for questions 11 to 16: Answer the questions on the basis of the information given below.

The table given below shows impartial information about the total number of tickets sold at a multiplex for four movie shows - A, B, C and D - at three Screens - I, II and III - during four time slots - 9 AM, 12 Noon, 3 PM and 6 PM - on a particular Sunday. The same movie is not Screened at the same time on any two Screens and every movie is shown at least once on each Screen. There are three types of seats for which tickets are sold - Box, Gold and Silver.

	9:00 AM	12 Noon	3:00 PM	6:00 PM	Total
Screen I	-	D 190	B 210	-	800
Screen II	C 180	-	-	-	840
Screen III	-	A 240	-	B	-
Total	-	-	-	760	

Further, it is also known that:
(i) Equal number of tickets is sold for all the 3 PM slots. The total number of tickets sold for the 9 AM slot was 200 less than that for the 6 PM slot. The total number of tickets sold for movies A and C were equal whereas for D the total was 10 more.
(ii) **For Screen I:** The total number of tickets sold for 9 AM to 6 PM time slots are in AP such that number of ticket sold under each category was also in AP. In the 6 PM slot 105 Gold tickets were sold whereas the Silver tickets sold were 5 more than the Box tickets, which was equal to the Silver tickets sold in the 3 PM slot. The Box tickets sold in the 9 AM slot of Screen I was 15 less than that in the 6 PM slot.
(iii) **For Screen II:** Equal number of tickets were sold in all three categories for the 9 AM as well as the 3 PM slots. The Box tickets for 12 Noon and 6 PM are 60 and 80 whereas the tickets sold in the remaining two categories for the two slots were equal.
(iv) **For Screen III:** The Box tickets sold for each time slot were equal to 60. The Silver tickets sold were in AP such that the number of 9 AM tickets were half the number of 6 PM tickets. The Gold and Silver tickets sold for the 6 PM slot were equal.

Q.11 [11831809]
Which of the following movies was shown on Screen II in the 6 PM slot?

---

1 ◯ A

---

2 ◯ D

---

3 ◯ B

4 ◯ Either (1) or (2)

Solution:
Correct Answer : 2

Answer key/Solution

Step 1: From condition (i),
We have to keep in mind that each movie is Screened once in each Screen and no two Screens show the same movie at the same time.
The movie in the 9 AM slot in Screen I cannot be C, so it has to be A.
For Screen III the movie in the 9 AM slot has to be D.
The movie in the 12 noon slot in Screen II has to be B.
The movie in the 3 PM slot in Screen III will be C.

Step 2:
For Screen I, the number of tickets sold for 9 AM to 6 PM time slots are in AP. So 170 tickets were sold for 9 AM and 230 tickets for 6 PM.
The total number of tickets sold for movies A and C were equal.
Let the tickets sold for A in the remaining slot be x.
Hence, (170 + 240 + x) = (180 + 210 + 230) ⇒ x = 210.
For D the total number was 620 + 10 = 630. So in the remaining slot for movie D, 230 tickets were sold. This means that D was not Screened at 3 pm in Screen II. So the 3 PM movie in Screen II was A and the 6 PM movie was D.
The total number of tickets sold in all the three Screens for all slots was 2600.
Let us complete the table with the above information:

	9:00 AM	12 noon	3:00 PM	6:00 PM	Total
Screen I	A 170	D 190	B 210	C 230	800
Screen II	C 180	B (840 - 180 - 210 - 230 = 220)	A 210	D 230	840
Screen III	D (560 - 180 - 170 = 210)	A 240	C 210	B (760 - 460 = 300)	960
Total	560	650	630	760	2600

Step 3:
Now let us find the tickets sold in each of the three categories for each time slot.
From condition (ii), **For Screen I:** The total number of tickets sold under each category was in AP.
In the 6 PM slot 105 Gold tickets were sold, so remaining = (230 – 105) also Silver tickets were 5 more than the Box tickets ⇒ Silver = 60 and Box = 65
Silver tickets sold in the 3 PM slot = 60. (So the common difference of this AP is 5).
Hence, the number of Silver tickets in other slots can be found.
The Box tickets sold in the 9 AM slot of Screen I was 15 less than that in the 6 PM slot. So the common difference of this AP is 5. So other values can be figured out. By subtracting these from the total we get the Gold tickets for each slot.
From condition (iii), **For Screen II:** Equal number of tickets were sold in all three categories for the 9 AM slot so the number of tickets for each category = 180/3 = 60 and for the 3 PM slots = 210/3 = 70. The Box tickets for 12 noon are 60, so the tickets sold in the remaining two categories = 80 and for the 6 PM slot 80 Box tickets are sold, so for the remaining two slots the tickets sold will be 75 each.
From condition (iv), **For Screen III:** The Box tickets sold for each time slot were equal to 60. The Gold and Silver tickets sold for the 6 PM slot were equal, so they will be (300 – 60)/2 = 120 each. The Silver tickets sold were in AP such that the number of 9 AM tickets were half the number of 6 PM tickets. So the number of 9 AM Silver tickets = 120/2 = 60. The number of Gold tickets for each slot can be calculated by subtracting the other two from the total for each slot.

		9 A.M.		12 Noon		3 P.M.		6 P.M.		total
Screen 1	Box	A	45	D	50	B	55	C	60	210
	Gold		75		85		95		105	360
	Silver	170	50	190	55	210	60	230	65	230
Screen 2	Box	C	60	B	60	A	70	D	80	270
	Gold		60		80		70		75	285
	Silver	180	60	220	80	210	70	230	75	285
Screen 3	Box	D	60	A	60	C	60	B	60	240
	Gold		90		100		50		120	360
	Silver	210	60	240	80	210	100	300	120	360

Movie D was shown in Screen II in the 6 PM slot.

Bookmark FeedBack

Directions for questions 11 to 16: Answer the questions on the basis of the information given below.

The table given below shows impartial information about the total number of tickets sold at a multiplex for four movie shows - A, B, C and D - at three Screens - I, II and III - during four time slots - 9 AM, 12 Noon, 3 PM and 6 PM - on a particular Sunday. The same movie is not Screened at the same time on any two Screens and every movie is shown at least once on each Screen. There are three types of seats for which tickets are sold - Box, Gold and Silver.

	9:00 AM	12 Noon	3:00 PM	6:00 PM	Total
Screen I	-	D 190	B 210	-	800
Screen II	C 180	-	-	-	840
Screen III	-	A 240	-	B	-
Total	-	-	-	760	

Further, it is also known that:
(i) Equal number of tickets is sold for all the 3 PM slots. The total number of tickets sold for the 9 AM slot was 200 less than that for the 6 PM slot. The total number of tickets sold for movies A and C were equal whereas for D the total was 10 more.
(ii) **For Screen I:** The total number of tickets sold for 9 AM to 6 PM time slots are in AP such that number of ticket sold under each category was also in AP. In the 6 PM slot 105 Gold tickets were sold whereas the Silver tickets sold were 5 more than the Box tickets, which was equal to the Silver tickets sold in the 3 PM slot. The Box tickets sold in the 9 AM slot of Screen I was 15 less than that in the 6 PM slot.
(iii) **For Screen II:** Equal number of tickets were sold in all three categories for the 9 AM as well as the 3 PM slots. The Box tickets for 12 Noon and 6 PM are 60 and 80 whereas the tickets sold in the remaining two categories for the two slots were equal.
(iv) **For Screen III:** The Box tickets sold for each time slot were equal to 60. The Silver tickets sold were in AP such that the number of 9 AM tickets were half the number of 6 PM tickets. The Gold and Silver tickets sold for the 6 PM slot were equal.

Q.12 [11831809]
Among all the 12 shows in the three Screens, how many shows had total ticket sales of more than 220?

1 ◯ Four

2 ◯ Five

3 ◯ Three

4 ◯ Six

Solution:
Correct Answer : 1

Answer key/Solution

Step 1: From condition (i),
We have to keep in mind that each movie is Screened once in each Screen and no two Screens show the same movie at the same time.
The movie in the 9 AM slot in Screen I cannot be C, so it has to be A.
For Screen III the movie in the 9 AM slot has to be D.
The movie in the 12 noon slot in Screen II has to be B.
The movie in the 3 PM slot in Screen III will be C.

Step 2:
For Screen I, the number of tickets sold for 9 AM to 6 PM time slots are in AP. So 170 tickets were sold for 9 AM and 230 tickets for 6 PM.
The total number of tickets sold for movies A and C were equal.
Let the tickets sold for A in the remaining slot be x.
Hence, (170 + 240 + x) = (180 + 210 + 230) ⇒ x = 210.
For D the total number was 620 + 10 = 630. So in the remaining slot for movie D, 230 tickets were sold. This means that D was not Screened at 3 pm in Screen II. So the 3 PM movie in Screen II was A and the 6 PM movie was D.
The total number of tickets sold in all the three Screens for all slots was 2600.
Let us complete the table with the above information:

	9:00 AM	12 noon	3:00 PM	6:00 PM	Total
Screen I	A 170	D 190	B 210	C 230	800
Screen II	C 180	B (840 - 180 - 210 - 230 = 220)	A 210	D 230	840
Screen III	D (560 - 180 - 170 = 210)	A 240	C 210	B (760 - 460 = 300)	960
Total	560	650	630	760	2600

Step 3:
Now let us find the tickets sold in each of the three categories for each time slot.
From condition (ii), **For Screen I:** The total number of tickets sold under each category was in AP.
In the 6 PM slot 105 Gold tickets were sold, so remaining = (230 – 105) also Silver tickets were 5 more than the Box tickets ⇒ Silver = 60 and Box = 65
Silver tickets sold in the 3 PM slot = 60. (So the common difference of this AP is 5).
Hence, the number of Silver tickets in other slots can be found.
The Box tickets sold in the 9 AM slot of Screen I was 15 less than that in the 6 PM slot. So the common difference of this AP is 5. So other values can be figured out. By subtracting these from the total we get the Gold tickets for each slot.
From condition (iii), **For Screen II:** Equal number of tickets were sold in all three categories for the 9 AM slot so the number of tickets for each category = 180/3 = 60 and for the 3 PM slots = 210/3 = 70. The Box tickets for 12 noon are 60, so the tickets sold in the remaining two categories = 80 and for the 6 PM slot 80 Box tickets are sold, so for the remaining two slots the tickets sold will be 75 each.
From condition (iv), **For Screen III:** The Box tickets sold for each time slot were equal to 60. The Gold and Silver tickets sold for the 6 PM slot were equal, so they will be (300 – 60)/2 = 120 each. The Silver tickets sold were in AP such that the number of 9 AM tickets were half the number of 6 PM tickets. So the number of 9 AM Silver tickets = 120/2 = 60. The number of Gold tickets for each slot can be calculated by subtracting the other two from the total for each slot.

		9 A.M.		12 Noon		3 P.M.		6 P.M.		total
Screen 1	Box	A	45	D	50	B	55	C	60	210
	Gold		75		85		95		105	360
	Silver	170	50	190	55	210	60	230	65	230
Screen 2	Box	C	60	B	60	A	70	D	80	270
	Gold		60		80		70		75	285
	Silver	180	60	220	80	210	70	230	75	285
Screen 3	Box	D	60	A	60	C	60	B	60	240
	Gold		90		100		50		120	360
	Silver	210	60	240	80	210	100	300	120	360

All the three 6 PM shows and the 12 noon show in Screen III had a total ticket sales of more than 220. Hence, the correct answer is four.

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Directions for questions 11 to 16: Answer the questions on the basis of the information given below.

The table given below shows impartial information about the total number of tickets sold at a multiplex for four movie shows - A, B, C and D - at three Screens - I, II and III - during four time slots - 9 AM, 12 Noon, 3 PM and 6 PM - on a particular Sunday. The same movie is not Screened at the same time on any two Screens and every movie is shown at least once on each Screen. There are three types of seats for which tickets are sold - Box, Gold and Silver.

	9:00 AM	12 Noon	3:00 PM	6:00 PM	Total
Screen I	-	D 190	B 210	-	800
Screen II	C 180	-	-	-	840
Screen III	-	A 240	-	B	-
Total	-	-	-	760	

Further, it is also known that:
(i) Equal number of tickets is sold for all the 3 PM slots. The total number of tickets sold for the 9 AM slot was 200 less than that for the 6 PM slot. The total number of tickets sold for movies A and C were equal whereas for D the total was 10 more.
(ii) **For Screen I:** The total number of tickets sold for 9 AM to 6 PM time slots are in AP such that number of ticket sold under each category was also in AP. In the 6 PM slot 105 Gold tickets were sold whereas the Silver tickets sold were 5 more than the Box tickets, which was equal to the Silver tickets sold in the 3 PM slot. The Box tickets sold in the 9 AM slot of Screen I was 15 less than that in the 6 PM slot.
(iii) **For Screen II:** Equal number of tickets were sold in all three categories for the 9 AM as well as the 3 PM slots. The Box tickets for 12 Noon and 6 PM are 60 and 80 whereas the tickets sold in the remaining two categories for the two slots were equal.
(iv) **For Screen III:** The Box tickets sold for each time slot were equal to 60. The Silver tickets sold were in AP such that the number of 9 AM tickets were half the number of 6 PM tickets. The Gold and Silver tickets sold for the 6 PM slot were equal.

Q.13 [11831809]
What percentage of the total tickets were sold for the 12 noon show in all three Screens together?

1 ◯ 20%

2 ◯ 25%

3 ◯ 30%

4 ◯ 18%

Solution:
Correct Answer : 2

Answer key/Solution

Step 1: From condition (i),
We have to keep in mind that each movie is Screened once in each Screen and no two Screens show the same movie at the same time.
The movie in the 9 AM slot in Screen I cannot be C, so it has to be A.
For Screen III the movie in the 9 AM slot has to be D.
The movie in the 12 noon slot in Screen II has to be B.
The movie in the 3 PM slot in Screen III will be C.

Step 2:
For Screen I, the number of tickets sold for 9 AM to 6 PM time slots are in AP. So 170 tickets were sold for 9 AM and 230 tickets for 6 PM.
The total number of tickets sold for movies A and C were equal.
Let the tickets sold for A in the remaining slot be x.
Hence, (170 + 240 + x) = (180 + 210 + 230) ⇒ x = 210.
For D the total number was 620 + 10 = 630. So in the remaining slot for movie D, 230 tickets were sold. This means that D was not Screened at 3 pm in Screen II. So the 3 PM movie in Screen II was A and the 6 PM movie was D.
The total number of tickets sold in all the three Screens for all slots was 2600.
Let us complete the table with the above information:

	9:00 AM	12 noon	3:00 PM	6:00 PM	Total
Screen I	A 170	D 190	B 210	C 230	800
Screen II	C 180	B (840 - 180 - 210 - 230 = 220)	A 210	D 230	840
Screen III	D (560 - 180 - 170 = 210)	A 240	C 210	B (760 - 460 = 300)	960
Total	560	650	630	760	2600

Step 3:
Now let us find the tickets sold in each of the three categories for each time slot.
From condition (ii), **For Screen I:** The total number of tickets sold under each category was in AP.
In the 6 PM slot 105 Gold tickets were sold, so remaining = (230 – 105) also Silver tickets were 5 more than the Box tickets ⇒ Silver = 60 and Box = 65
Silver tickets sold in the 3 PM slot = 60. (So the common difference of this AP is 5).
Hence, the number of Silver tickets in other slots can be found.
The Box tickets sold in the 9 AM slot of Screen I was 15 less than that in the 6 PM slot. So the common difference of this AP is 5. So other values can be figured out. By subtracting these from the total we get the Gold tickets for each slot.
From condition (iii), **For Screen II:** Equal number of tickets were sold in all three categories for the 9 AM slot so the number of tickets for each category = 180/3 = 60 and for the 3 PM slots = 210/3 = 70. The Box tickets for 12 noon are 60, so the tickets sold in the remaining two categories = 80 and for the 6 PM slot 80 Box tickets are sold, so for the remaining two slots the tickets sold will be 75 each.
From condition (iv), **For Screen III:** The Box tickets sold for each time slot were equal to 60. The Gold and Silver tickets sold for the 6 PM slot were equal, so they will be (300 – 60)/2 = 120 each. The Silver tickets sold were in AP such that the number of 9 AM tickets were half the number of 6 PM tickets. So the number of 9 AM Silver tickets = 120/2 = 60. The number of Gold tickets for each slot can be calculated by subtracting the other two from the total for each slot.

		9 A.M.		12 Noon		3 P.M.		6 P.M.		total
Screen 1	Box	A	45	D	50	B	55	C	60	210
	Gold		75		85		95		105	360
	Silver	170	50	190	55	210	60	230	65	230
Screen 2	Box	C	60	B	60	A	70	D	80	270
	Gold		60		80		70		75	285
	Silver	180	60	220	80	210	70	230	75	285
Screen 3	Box	D	60	A	60	C	60	B	60	240
	Gold		90		100		50		120	360
	Silver	210	60	240	80	210	100	300	120	360

Required percentage = 650/2600 × 100 = 25%.

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Directions for questions 11 to 16: Answer the questions on the basis of the information given below.

The table given below shows impartial information about the total number of tickets sold at a multiplex for four movie shows - A, B, C and D - at three Screens - I, II and III - during four time slots - 9 AM, 12 Noon, 3 PM and 6 PM - on a particular Sunday. The same movie is not Screened at the same time on any two Screens and every movie is shown at least once on each Screen. There are three types of seats for which tickets are sold - Box, Gold and Silver.

	9:00 AM	12 Noon	3:00 PM	6:00 PM	Total
Screen I	-	D 190	B 210	-	800
Screen II	C 180	-	-	-	840
Screen III	-	A 240	-	B	-
Total	-	-	-	760	

Further, it is also known that:
(i) Equal number of tickets is sold for all the 3 PM slots. The total number of tickets sold for the 9 AM slot was 200 less than that for the 6 PM slot. The total number of tickets sold for movies A and C were equal whereas for D the total was 10 more.
(ii) **For Screen I:** The total number of tickets sold for 9 AM to 6 PM time slots are in AP such that number of ticket sold under each category was also in AP. In the 6 PM slot 105 Gold tickets were sold whereas the Silver tickets sold were 5 more than the Box tickets, which was equal to the Silver tickets sold in the 3 PM slot. The Box tickets sold in the 9 AM slot of Screen I was 15 less than that in the 6 PM slot.
(iii) **For Screen II:** Equal number of tickets were sold in all three categories for the 9 AM as well as the 3 PM slots. The Box tickets for 12 Noon and 6 PM are 60 and 80 whereas the tickets sold in the remaining two categories for the two slots were equal.
(iv) **For Screen III:** The Box tickets sold for each time slot were equal to 60. The Silver tickets sold were in AP such that the number of 9 AM tickets were half the number of 6 PM tickets. The Gold and Silver tickets sold for the 6 PM slot were equal.

Q.14 [11831809]
What was the total number of Box tickets sold for movie D?

1 ◯ 180

2 ◯ 220

3 ◯ 190

4 ◯ 240

Solution:
Correct Answer : 3

Answer key/Solution

Step 1: From condition (i),
We have to keep in mind that each movie is Screened once in each Screen and no two Screens show the same movie at the same time.
The movie in the 9 AM slot in Screen I cannot be C, so it has to be A.
For Screen III the movie in the 9 AM slot has to be D.
The movie in the 12 noon slot in Screen II has to be B.
The movie in the 3 PM slot in Screen III will be C.

Step 2:
For Screen I, the number of tickets sold for 9 AM to 6 PM time slots are in AP. So 170 tickets were sold for 9 AM and 230 tickets for 6 PM.
The total number of tickets sold for movies A and C were equal.
Let the tickets sold for A in the remaining slot be x.
Hence, (170 + 240 + x) = (180 + 210 + 230) ⇒ x = 210.
For D the total number was 620 + 10 = 630. So in the remaining slot for movie D, 230 tickets were sold. This means that D was not Screened at 3 pm in Screen II. So the 3 PM movie in Screen II was A and the 6 PM movie was D.
The total number of tickets sold in all the three Screens for all slots was 2600.
Let us complete the table with the above information:

	9:00 AM	12 noon	3:00 PM	6:00 PM	Total
Screen I	A 170	D 190	B 210	C 230	800
Screen II	C 180	B (840 - 180 - 210 - 230 = 220)	A 210	D 230	840
Screen III	D (560 - 180 - 170 = 210)	A 240	C 210	B (760 - 460 = 300)	960
Total	560	650	630	760	2600

Step 3:
Now let us find the tickets sold in each of the three categories for each time slot.
From condition (ii), **For Screen I:** The total number of tickets sold under each category was in AP.
In the 6 PM slot 105 Gold tickets were sold, so remaining = (230 – 105) also Silver tickets were 5 more than the Box tickets ⇒ Silver = 60 and Box = 65
Silver tickets sold in the 3 PM slot = 60. (So the common difference of this AP is 5).
Hence, the number of Silver tickets in other slots can be found.
The Box tickets sold in the 9 AM slot of Screen I was 15 less than that in the 6 PM slot. So the common difference of this AP is 5. So other values can be figured out. By subtracting these from the total we get the Gold tickets for each slot.
From condition (iii), **For Screen II:** Equal number of tickets were sold in all three categories for the 9 AM slot so the number of tickets for each category = 180/3 = 60 and for the 3 PM slots = 210/3 = 70. The Box tickets for 12 noon are 60, so the tickets sold in the remaining two categories = 80 and for the 6 PM slot 80 Box tickets are sold, so for the remaining two slots the tickets sold will be 75 each.
From condition (iv), **For Screen III:** The Box tickets sold for each time slot were equal to 60. The Gold and Silver tickets sold for the 6 PM slot were equal, so they will be (300 – 60)/2 = 120 each. The Silver tickets sold were in AP such that the number of 9 AM tickets were half the number of 6 PM tickets. So the number of 9 AM Silver tickets = 120/2 = 60. The number of Gold tickets for each slot can be calculated by subtracting the other two from the total for each slot.

		9 A.M.		12 Noon		3 P.M.		6 P.M.		total
Screen 1	Box	A	45	D	50	B	55	C	60	210
	Gold		75		85		95		105	360
	Silver	170	50	190	55	210	60	230	65	230
Screen 2	Box	C	60	B	60	A	70	D	80	270
	Gold		60		80		70		75	285
	Silver	180	60	220	80	210	70	230	75	285
Screen 3	Box	D	60	A	60	C	60	B	60	240
	Gold		90		100		50		120	360
	Silver	210	60	240	80	210	100	300	120	360

Total number of Box tickets for D = 50 + 80 + 60 = 190.

Bookmark FeedBack

Directions for questions 11 to 16: Answer the questions on the basis of the information given below.

The table given below shows impartial information about the total number of tickets sold at a multiplex for four movie shows - A, B, C and D - at three Screens - I, II and III - during four time slots - 9 AM, 12 Noon, 3 PM and 6 PM - on a particular Sunday. The same movie is not Screened at the same time on any two Screens and every movie is shown at least once on each Screen. There are three types of seats for which tickets are sold - Box, Gold and Silver.

	9:00 AM	12 Noon	3:00 PM	6:00 PM	Total
Screen I	-	D 190	B 210	-	800
Screen II	C 180	-	-	-	840
Screen III	-	A 240	-	B	-
Total	-	-	-	760	

Further, it is also known that:
(i) Equal number of tickets is sold for all the 3 PM slots. The total number of tickets sold for the 9 AM slot was 200 less than that for the 6 PM slot. The total number of tickets sold for movies A and C were equal whereas for D the total was 10 more.
(ii) **For Screen I:** The total number of tickets sold for 9 AM to 6 PM time slots are in AP such that number of ticket sold under each category was also in AP. In the 6 PM slot 105 Gold tickets were sold whereas the Silver tickets sold were 5 more than the Box tickets, which was equal to the Silver tickets sold in the 3 PM slot. The Box tickets sold in the 9 AM slot of Screen I was 15 less than that in the 6 PM slot.
(iii) **For Screen II:** Equal number of tickets were sold in all three categories for the 9 AM as well as the 3 PM slots. The Box tickets for 12 Noon and 6 PM are 60 and 80 whereas the tickets sold in the remaining two categories for the two slots were equal.
(iv) **For Screen III:** The Box tickets sold for each time slot were equal to 60. The Silver tickets sold were in AP such that the number of 9 AM tickets were half the number of 6 PM tickets. The Gold and Silver tickets sold for the 6 PM slot were equal.

Q.15 [11831809]
What is the difference between the Gold tickets sold in Screen I and the Gold tickets sold in Screen III?

Solution:

Correct Answer : 0

Answer key/Solution

Step 1: From condition (i),
We have to keep in mind that each movie is Screened once in each Screen and no two Screens show the same movie at the same time.
The movie in the 9 AM slot in Screen I cannot be C, so it has to be A.
For Screen III the movie in the 9 AM slot has to be D.
The movie in the 12 noon slot in Screen II has to be B.
The movie in the 3 PM slot in Screen III will be C.

Step 2:
For Screen I, the number of tickets sold for 9 AM to 6 PM time slots are in AP. So 170 tickets were sold for 9 AM and 230 tickets for 6 PM.
The total number of tickets sold for movies A and C were equal.
Let the tickets sold for A in the remaining slot be x.
Hence, (170 + 240 + x) = (180 + 210 + 230) ⇒ x = 210.
For D the total number was 620 + 10 = 630. So in the remaining slot for movie D, 230 tickets were sold. This means that D was not Screened at 3 pm in Screen II. So the 3 PM movie in Screen II was A and the 6 PM movie was D.
The total number of tickets sold in all the three Screens for all slots was 2600.
Let us complete the table with the above information:

	9:00 AM	12 noon	3:00 PM	6:00 PM	Total
Screen I	A 170	D 190	B 210	C 230	800
Screen II	C 180	B (840 - 180 - 210 - 230 = 220)	A 210	D 230	840
Screen III	D (560 - 180 - 170 = 210)	A 240	C 210	B (760 - 460 = 300)	960
Total	560	650	630	760	2600

Step 3:
Now let us find the tickets sold in each of the three categories for each time slot.
From condition (ii), **For Screen I:** The total number of tickets sold under each category was in AP.
In the 6 PM slot 105 Gold tickets were sold, so remaining = (230 – 105) also Silver tickets were 5 more than the Box tickets ⇒ Silver = 60 and Box = 65
Silver tickets sold in the 3 PM slot = 60. (So the common difference of this AP is 5).
Hence, the number of Silver tickets in other slots can be found.
The Box tickets sold in the 9 AM slot of Screen I was 15 less than that in the 6 PM slot. So the common difference of this AP is 5. So other values can be figured out. By subtracting these from the total we get the Gold tickets for each slot.
From condition (iii), **For Screen II:** Equal number of tickets were sold in all three categories for the 9 AM slot so the number of tickets for each category = 180/3 = 60 and for the 3 PM slots = 210/3 = 70. The Box tickets for 12 noon are 60, so the tickets sold in the remaining two categories = 80 and for the 6 PM slot 80 Box tickets are sold, so for the remaining two slots the tickets sold will be 75 each.
From condition (iv), **For Screen III:** The Box tickets sold for each time slot were equal to 60. The Gold and Silver tickets sold for the 6 PM slot were equal, so they will be (300 – 60)/2 = 120 each. The Silver tickets sold were in AP such that the number of 9 AM tickets were half the number of 6 PM tickets. So the number of 9 AM Silver tickets = 120/2 = 60. The number of Gold tickets for each slot can be calculated by subtracting the other two from the total for each slot.

		9 A.M.		12 Noon		3 P.M.		6 P.M.		total
Screen 1	Box	A	45	D	50	B	55	C	60	210
	Gold		75		85		95		105	360
	Silver	170	50	190	55	210	60	230	65	230
Screen 2	Box	C	60	B	60	A	70	D	80	270
	Gold		60		80		70		75	285
	Silver	180	60	220	80	210	70	230	75	285
Screen 3	Box	D	60	A	60	C	60	B	60	240
	Gold		90		100		50		120	360
	Silver	210	60	240	80	210	100	300	120	360

Gold tickets in Screen I = 360
Gold tickets in Screen III = 360
Hence, require difference = 0.

Bookmark FeedBack

Directions for questions 11 to 16: Answer the questions on the basis of the information given below.

The table given below shows impartial information about the total number of tickets sold at a multiplex for four movie shows - A, B, C and D - at three Screens - I, II and III - during four time slots - 9 AM, 12 Noon, 3 PM and 6 PM - on a particular Sunday. The same movie is not Screened at the same time on any two Screens and every movie is shown at least once on each Screen. There are three types of seats for which tickets are sold - Box, Gold and Silver.

	9:00 AM	12 Noon	3:00 PM	6:00 PM	Total
Screen I	-	D 190	B 210	-	800
Screen II	C 180	-	-	-	840
Screen III	-	A 240	-	B	-
Total	-	-	-	760	

Further, it is also known that:
(i) Equal number of tickets is sold for all the 3 PM slots. The total number of tickets sold for the 9 AM slot was 200 less than that for the 6 PM slot. The total number of tickets sold for movies A and C were equal whereas for D the total was 10 more.
(ii) **For Screen I:** The total number of tickets sold for 9 AM to 6 PM time slots are in AP such that number of ticket sold under each category was also in AP. In the 6 PM slot 105 Gold tickets were sold whereas the Silver tickets sold were 5 more than the Box tickets, which was equal to the Silver tickets sold in the 3 PM slot. The Box tickets sold in the 9 AM slot of Screen I was 15 less than that in the 6 PM slot.
(iii) **For Screen II:** Equal number of tickets were sold in all three categories for the 9 AM as well as the 3 PM slots. The Box tickets for 12 Noon and 6 PM are 60 and 80 whereas the tickets sold in the remaining two categories for the two slots were equal.
(iv) **For Screen III:** The Box tickets sold for each time slot were equal to 60. The Silver tickets sold were in AP such that the number of 9 AM tickets were half the number of 6 PM tickets. The Gold and Silver tickets sold for the 6 PM slot were equal.

Q.16 [11831809]
What is the ratio of the total number of Box tickets and Silver tickets respectively sold in the 12 noon slot in all three Screens together?

1 ◯ 33 : 43

2 ◯ 37 : 51

3 ◯ 43 : 57

4 ◯ 34 : 43

Solution:

Correct Answer : 4

Answer key/Solution

Step 1: From condition (i),

We have to keep in mind that each movie is Screened once in each Screen and no two Screens show the same movie at the same time.

The movie in the 9 AM slot in Screen I cannot be C, so it has to be A.

For Screen III the movie in the 9 AM slot has to be D.

The movie in the 12 noon slot in Screen II has to be B.

The movie in the 3 PM slot in Screen III will be C.

Step 2:

For Screen I, the number of tickets sold for 9 AM to 6 PM time slots are in AP. So 170 tickets were sold for 9 AM and 230 tickets for 6 PM.

The total number of tickets sold for movies A and C were equal.

Let the tickets sold for A in the remaining slot be x.

Hence, (170 + 240 + x) = (180 + 210 + 230) ⇒ x = 210.

For D the total number was 620 + 10 = 630. So in the remaining slot for movie D, 230 tickets were sold. This means that D was not Screened at 3 pm in Screen II. So the 3 PM movie in Screen II was A and the 6 PM movie was D.

The total number of tickets sold in all the three Screens for all slots was 2600.

Let us complete the table with the above information:

	9:00 AM	12 noon	3:00 PM	6:00 PM	Total
Screen I	A 170	D 190	B 210	C 230	800
Screen II	C 180	B (840 - 180 - 210 - 230 = 220)	A 210	D 230	840
Screen III	D (560 - 180 - 170 = 210)	A 240	C 210	B (760 - 460 = 300)	960
Total	560	650	630	760	2600

Step 3:

Now let us find the tickets sold in each of the three categories for each time slot.

From condition (ii), **For Screen I:** The total number of tickets sold under each category was in AP.

In the 6 PM slot 105 Gold tickets were sold, so remaining = (230 – 105) also Silver tickets were 5 more than the Box tickets ⇒ Silver = 60 and Box = 65

Silver tickets sold in the 3 PM slot = 60. (So the common difference of this AP is 5).

Hence, the number of Silver tickets in other slots can be found.

The Box tickets sold in the 9 AM slot of Screen I was 15 less than that in the 6 PM slot. So the common difference of this AP is 5. So other values can be figured out. By subtracting these from the total we get the Gold tickets for each slot.

From condition (iii), **For Screen II:** Equal number of tickets were sold in all three categories for the 9 AM slot so the number of tickets for each category = 180/3 = 60 and for the 3 PM slots = 210/3 = 70. The Box tickets for 12 noon are 60, so the tickets sold in the remaining two categories = 80 and for the 6 PM slot 80 Box tickets are sold, so for the remaining two slots the tickets sold will be 75 each.

From condition (iv), **For Screen III:** The Box tickets sold for each time slot were equal to 60. The Gold and Silver tickets sold for the 6 PM slot were equal, so they will be (300 – 60)/2 = 120 each. The Silver tickets sold were in AP such that the number of 9 AM tickets were half the number of 6 PM tickets. So the number of 9 AM Silver tickets = 120/2 = 60. The number of Gold tickets for each slot can be calculated by subtracting the other two from the total for each slot.

		9 A.M.		12 Noon		3 P.M.		6 P.M.		total
Screen 1	Box	A	45	D	50	B	55	C	60	210
	Gold		75		85		95		105	360
	Silver	170	50	190	55	210	60	230	65	230
Screen 2	Box	C	60	B	60	A	70	D	80	270
	Gold		60		80		70		75	285
	Silver	180	60	220	80	210	70	230	75	285
Screen 3	Box	D	60	A	60	C	60	B	60	240
	Gold		90		100		50		120	360
	Silver	210	60	240	80	210	100	300	120	360

Box tickets sold in the 12 noon slot in all three Screens = 50 + 60 + 60 = 170
Silver tickets sold in the 12 noon slot in all three Screens = 55 + 80 + 80 = 215
Required ratio = 170 : 215 = 34 : 43.

Bookmark FeedBack

Directions for questions 17 to 20: Answer the questions on the basis of the information given below.

Eight cabinet ministers – A, B, C, D, E, F, G and H are members of at least one of the following three cabinet committees: Economic Affairs, Parliamentary Affairs and Political Affairs. Among them two are members of Economic Affairs but not of Parliamentary Affairs or Political Affairs, another two are members of Parliamentary Affairs but not of Economic Affairs or Political Affairs, and one is a member of Political Affairs but not of Parliamentary Affairs or Economic Affairs. Further, two are members of Economic Affairs and Parliamentary Affairs but not of Political Affairs, and one is a member of Economic Affairs and Political Affairs but not of Parliamentary Affairs.
The following facts are known about these eight cabinet ministers.
(i) Out of A and B, one belongs to the Committee on Economic Affairs only and the other belongs to the Committee on Parliamentary Affairs only.
(ii) E is a member of both Economic Affairs and Political Affairs.
(iii) Neither F nor H is a member of Parliamentary Affairs.
(iv) Out of C and D, exactly one belongs to the Economic Affairs Committee.

Q.17 [11831809]
How many of the ministers cannot be a possible member of the committee on Economic Affairs?

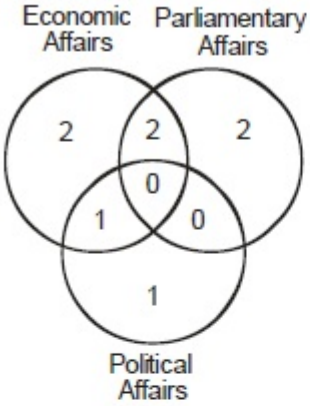
Solution:

Correct Answer : 0

Answer key/Solution

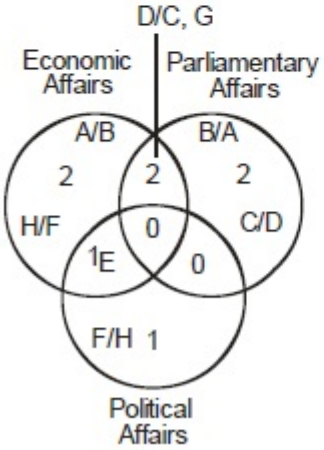
Step 1:

From the given information, the diagram can be shown as:



Step 2:

From conditions (i) to (iv), the diagram can be shown as:



All ministers can be a possible member of the committee on Economic Affairs.

Bookmark FeedBack

Directions for questions 17 to 20: Answer the questions on the basis of the information given below.

Eight cabinet ministers – A, B, C, D, E, F, G and H are members of at least one of the following three cabinet committees: Economic Affairs, Parliamentary Affairs and Political Affairs. Among them two are members of Economic Affairs but not of Parliamentary Affairs or Political Affairs, another two are members of Parliamentary Affairs but not of Economic Affairs or Political Affairs, and one is a member of Political Affairs but not of Parliamentary Affairs or Economic Affairs. Further, two are members of Economic Affairs and Parliamentary Affairs but not of Political Affairs, and one is a member of Economic Affairs and Political Affairs but not of Parliamentary Affairs.
The following facts are known about these eight cabinet ministers.
(i) Out of A and B, one belongs to the Committee on Economic Affairs only and the other belongs to the Committee on Parliamentary Affairs only.
(ii) E is a member of both Economic Affairs and Political Affairs.
(iii) Neither F nor H is a member of Parliamentary Affairs.
(iv) Out of C and D, exactly one belongs to the Economic Affairs Committee.

Q.18 [11831809]
Who among the following is definitely a member of both the Economic Affairs and Parliamentary Affairs Committees?

1 ◯ C

2 ◯ G

3 ◯ D

4 ◯ F

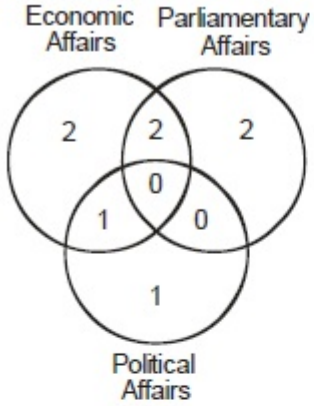
Solution:

Correct Answer : 2

Answer key/Solution

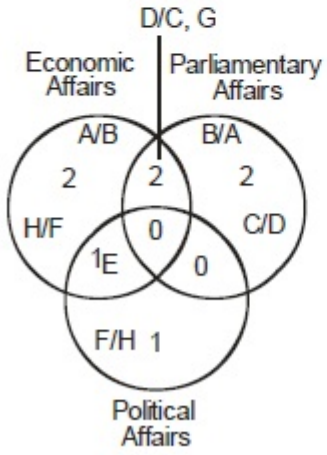
Step 1:

From the given information, the diagram can be shown as:



Step 2:

From conditions (i) to (iv), the diagram can be shown as:



G is definitely a member of both Economic Affairs and Parliamentary Affairs.

Bookmark FeedBack

Directions for questions 17 to 20: Answer the questions on the basis of the information given below.

Eight cabinet ministers – A, B, C, D, E, F, G and H are members of at least one of the following three cabinet committees: Economic Affairs, Parliamentary Affairs and Political Affairs. Among them two are members of Economic Affairs but not of Parliamentary Affairs or Political Affairs, another two are members of Parliamentary Affairs but not of Economic Affairs or Political Affairs, and one is a member of Political Affairs but not of Parliamentary Affairs or Economic Affairs. Further, two are members of Economic Affairs and Parliamentary Affairs but not of Political Affairs, and one is a member of Economic Affairs and Political Affairs but not of Parliamentary Affairs.
The following facts are known about these eight cabinet ministers.
(i) Out of A and B, one belongs to the Committee on Economic Affairs only and the other belongs to the Committee on Parliamentary Affairs only.
(ii) E is a member of both Economic Affairs and Political Affairs.
(iii) Neither F nor H is a member of Parliamentary Affairs.
(iv) Out of C and D, exactly one belongs to the Economic Affairs Committee.

Q.19 [11831809]
If A, C and F are members of different committees, then which of the following pairs belongs to the same committee?

1 ◯ C and H

2 ◯ B and F

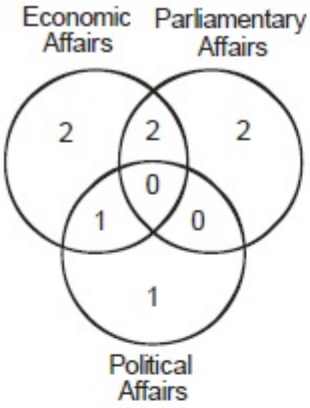
3 ◯ B and E

4 ◯ B and C

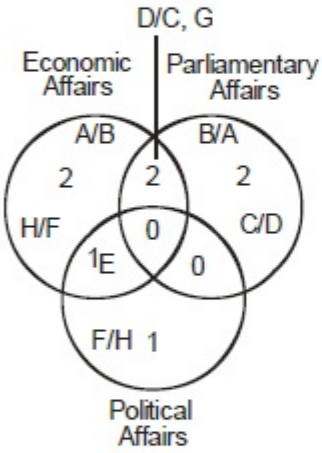
Solution:
Correct Answer : 4

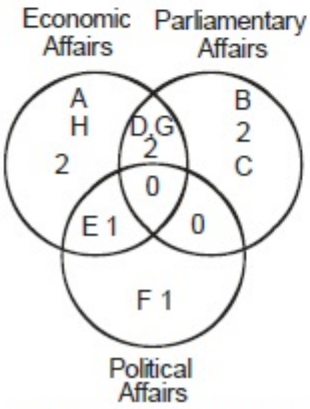
Answer key/Solution

Step 1:
From the given information, the diagram can be shown as:



Step 2:
From conditions (i) to (iv), the diagram can be shown as:





If A, C and F are members of different committees, then B and C are members of the Parliamentary Affairs Committee

Bookmark FeedBack

Directions for questions 17 to 20: Answer the questions on the basis of the information given below.

Eight cabinet ministers – A, B, C, D, E, F, G and H are members of at least one of the following three cabinet committees: Economic Affairs, Parliamentary Affairs and Political Affairs. Among them two are members of Economic Affairs but not of Parliamentary Affairs or Political Affairs, another two are members of Parliamentary Affairs but not of Economic Affairs or Political Affairs, and one is a member of Political Affairs but not of Parliamentary Affairs or Economic Affairs. Further, two are members of Economic Affairs and Parliamentary Affairs but not of Political Affairs, and one is a member of Economic Affairs and Political Affairs but not of Parliamentary Affairs.
The following facts are known about these eight cabinet ministers.
(i) Out of A and B, one belongs to the Committee on Economic Affairs only and the other belongs to the Committee on Parliamentary Affairs only.
(ii) E is a member of both Economic Affairs and Political Affairs.
(iii) Neither F nor H is a member of Parliamentary Affairs.
(iv) Out of C and D, exactly one belongs to the Economic Affairs Committee.

Q.20 [11831809]
If D and F cannot work together in a committee, then which of the following statements can be FALSE?

1 ◯ D and G can be members of both the Economic Affairs and Parliamentary Affairs committees.

2 ◯ D and H can be members of the Economic Affairs committee.

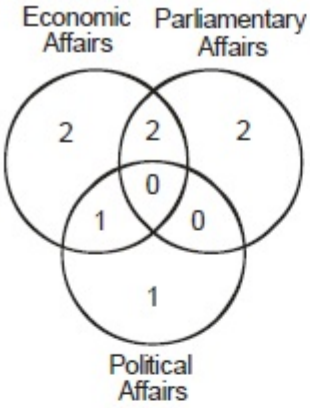
3 ◯ B, C and H cannot be the members of the same committee.

4 ◯ A and F can work together in the same committee.

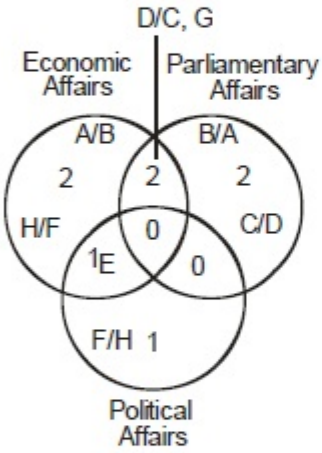
Solution:
Correct Answer : 3

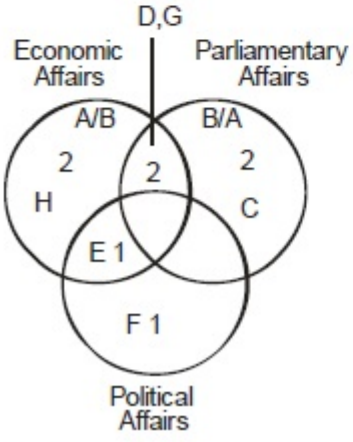
Answer key/Solution

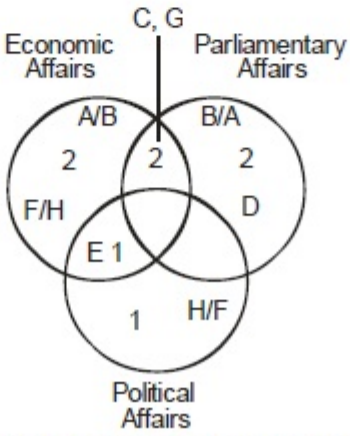
Step 1:
From the given information, the diagram can be shown as:



Step 2:
From conditions (i) to (iv), the diagram can be shown as:







If D and F cannot work together in a committee, the statement given in option (3) can be FALSE.

Bookmark FeedBack